Real-world — a support-reply router — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried.

- Hint 1 — One reply chain per category, then branch
- Hint 2 — The plan, and how to prove routing

Hint 1 — One reply chain per category, then branch

Keep each part in its own small function so the pipeline reads like a sentence:

- `build_reply_chain(model, system_text) -> Runnable` — one category's chain.
- `build_router(model) -> Runnable` — the `RunnableBranch` over all categories.
- `build_pipeline(model) -> Runnable` — `build_router(model) | RunnableLambda(tag_reply)`.
- The default (last) argument to `RunnableBranch` has **no condition** — it's what runs when nothing matched. Don't forget it, or an unknown category raises.

Hint 2 — The plan, and how to prove routing

To prove routing **without** spending a token, give each branch a tiny fake model that echoes the system line it was handed. Then the reply tells you which branch ran:

```
fake model: return AIMessage(content = the system message it received)
test 1: billing -> reply mentions "billing agent";
        technical -> "tech-support agent"; sales -> "general support agent"
test 2: an unknown category -> the default branch, and "(auto-routed)" is there
```

Because the branch condition is on `ticket["category"]`, not on the model's words, the fake model is enough to prove the routing is correct. Then run `main()` once with a real `ChatOpenAI` to see real replies.

Full solution: Show me the solution